

MD AL WALID

Working Student — Full-Stack Web Development

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PROFILE

Full-stack developer with a background in Information & Communication Engineering, currently pursuing a Master's degree in Web Engineering at TU Chemnitz. Nearly two years of professional experience as an Associate in an R&D team at a multinational company, developing strong problem-solving skills within a structured, collaborative engineering environment. Through hands-on projects, I have built full-stack web applications using modern technologies including React, Node.js, Express, and PostgreSQL/MongoDB. Seeking a working student (Werkstudent) position in web development to apply this experience alongside my studies, keep growing as a developer, and contribute to a team that values good engineering practices and continuous learning.

WORK EXPERIENCE

Associate — Research & Development Team

Quantanite (Multinational Company) | Dec 2022 – Sep 2024 | Dhaka, Bangladesh

- Managed and maintained large-scale datasets to support AI training initiatives, ensuring high levels of data accuracy, consistency, and quality.
- Prepared and processed raw data according to project requirements while maintaining defined quality standards.
- Identified workflow challenges and implemented process improvements to increase efficiency, productivity, and overall output quality.
- Collaborated with internal teams and communicated directly with clients to understand requirements, resolve issues, and provide effective solutions.
- Trained and supported new team members by sharing process knowledge, quality guidelines, and best practices.
- Adapted workflows based on evolving business requirements while maintaining accuracy and timely delivery.

Intern Trainee — Network Operations Center

Bangladesh Submarine Cable Company Ltd. (BSCCL) | Feb 2020 – Mar 2020 | Kuakata, Bangladesh

- Set up hardware and software configurations in the Network Operations Center to meet performance requirements.
- Helped formulate data center expenditure reports for management.

Industrial Trainee

Western Marine Shipyard Limited | Aug 2016 – Nov 2016 | Chattogram, Bangladesh

- Gained practical experience in tool management and CNC machine operations.
- Monitored shipboard repair work to meet quality, safety, and efficiency standards.

FULL-STACK PROJECTS

Full-Stack E-Commerce Platform

Personal Project

<https://ecommerce-website-delta.vercel.app/>

- Built a full-stack e-commerce platform with customer and admin roles, including registration, login, JWT authentication, and role-based authorization.
- Designed a normalized PostgreSQL schema with relationships, constraints, and indexes to support products, categories, carts, and orders.
- Developed a RESTful API with Node.js and Express, including product search, filtering, sorting, and pagination, request validation, and centralized error handling.
- Implemented shopping cart, checkout, order creation, and customer order history, plus an admin dashboard with platform statistics, product and category CRUD, and order status management.
- Containerized the application with Docker and set up GitHub Actions CI/CD pipelines for the frontend and backend, with automatic frontend deployment to Vercel.
- Tech stack: React, Bootstrap, Axios, Node.js, Express, PostgreSQL, JWT, bcrypt, Docker, GitHub Actions.

German Traffic Accident Analytics System (ETL Data Hub — Full-Stack Data Pipeline)

Personal Project | 06/2026 – 09/2026

<https://german-traffic-data.vercel.app/>

- Built a full-stack ETL data pipeline extracting data from multiple public sources (CSV and ZIP archives) and loading it into a normalized PostgreSQL database.

- Designed and implemented data transformation logic (parsing, unpivoting, and cleaning) to convert raw datasets into structured, query-ready formats.
- Developed a RESTful API using Node.js and Express, exposing dedicated endpoints per dataset.
- Built a responsive React (Vite) dashboard with Bootstrap, featuring dataset selection, loading states, and error handling.
- Tech stack: Node.js, Express, PostgreSQL, React, Bootstrap, Axios, csv-parser, adm-zip.

Notes App — Full-Stack Web Application

Personal Project

<https://note-clener.vercel.app/>

- Developed a full-stack notes management application with user authentication and CRUD functionality.
- Implemented JWT authentication and bcrypt password hashing with protected frontend and backend routes.
- Built RESTful APIs for creating, reading, updating, and deleting user-specific notes.
- Used React Context API and Axios interceptors to manage authentication state and automatically attach JWT tokens to API requests.
- Implemented middleware for authentication, authorization, CORS, validation, and centralized error handling.
- Dockerized React frontend & Express backend using Docker Compose and multi-stage Docker build with Nginx.
- Deployed the frontend on Vercel and backend API on Render with environment-based configuration.

Zerodha Simple Clone — Full-Stack Trading Platform

Personal Project

<https://zerodha-website-ecru.vercel.app/>

- Built a full-stack, Zerodha-inspired stock trading platform using React, Node.js, Express.js, and MongoDB.
- Developed core trading features including holdings management, watchlist management, portfolio tracking, and a buy/sell order interface.
- Designed and integrated RESTful APIs (e.g. /allHoldings, /allPositions, /newOrder, /watchlist) to connect frontend and backend.
- Built a responsive landing page and interactive trading dashboard using React and Bootstrap, optimized for both desktop and mobile.
- Tech stack: React, Bootstrap, Axios, React Router, Node.js, Express.js, MongoDB.

Collaborative-Filtering Recommender System

Personal Project

- Designed an adaptive question-recommendation system using cosine similarity and Pearson correlation, built and analyzed in Jupyter Notebook (Python).

SKILLS

Frontend: HTML, CSS, JavaScript, React, Bootstrap

Backend: Node.js, Express.js, RESTful API Design (CRUD Operations), Swagger/OpenAPI Documentation

Databases: PostgreSQL, MongoDB (used in projects), MySQL (basic)

Data Management: Data Accuracy & Quality Assurance, Large-Scale Dataset Management, AI Training Data Preparation

Programming Languages: C, C++, Java, Python (basic)

Soft Skills: Cross-Functional Collaboration, Client Communication, Process Improvement, Adaptability, Mentoring

EDUCATION

M.Sc. in Web Engineering

Technische Universität Chemnitz | Oct 2024 – Present | Chemnitz, Germany

B.Sc. in Information & Communication Engineering

Bangladesh Army University of Engineering & Technology (BAUET) | 2017 – 2022 | Natore, Bangladesh

Diploma in Marine Technology

City Engineering Institute | 2012 – 2017 | Khulna, Bangladesh

LANGUAGES

Bengali: Mother tongue

English: B2 (Independent user) — Listening, Reading, Spoken, Writing

German: A2 (Basic user)

ADDITIONAL

Organizing Secretary, BAUET ICT Club — coordinated club events and member communication (2021–2022)